



# JavaScript Functions - Real World Mini Project Assignments

## 🔧!! Submission Guidelines

- Use **HTML + Tailwind CSS + JavaScript** only.
- Do not use `addEventListener`; use `onclick` in HTML.
- Use **modular functions** and avoid inline logic.
- Every project should be in a separate folder named: with project relavent names.
- Push all folders to a **single GitHub repo** named: `js-mini-projects`
- Deadline: 13/07/2025

### 1. Discount Calculator (Default Parameters)

Create a simple price calculator that optionally includes tax. If no tax is provided, a default value should be used.

- Input: Product price
- Optional: Tax (default 5%)
- Output: Final price after tax
- Concept: Default parameters, return statement

### 2. Live Clock

Display a digital clock that updates every second, showing the current system time.

- Show system time in real-time
- Update every second
- Concept: Function Expression, `setInterval`, return

### 3. Countdown Timer (IIFE)

Automatically start a countdown when the page loads to display how much time is left for a special offer.

- Auto-start countdown to offer expiry (e.g., 3 days left)
- Concept: IIFE, DOM, return, `setInterval`

### 4. Start/Stop Timer (Callback)

Allow users to start and stop a timer that counts how long a session has lasted.

- Show seconds passed
- Use callback to update DOM
- Concept: Callback function, `setInterval`, clear interval

### 5. Number Highlighter with Recursion

Automatically display numbers in sequence with a 1-second delay between each number, using recursion.

- Input: Range (e.g., 1 to 10)
- Highlight numbers one by one every second
- Concept: Recursive function, base case, `setTimeout`

### 6. Star Rating Renderer

Generate 5 stars using a loop and allow users to click and rate something visually. Show rating score.

- Dynamically render 5 stars using a loop
- On click, highlight selected rating and show score
- Concept: DOM rendering with recursion or dynamic generation

### 7. Filter Product Prices (HOF)

Let users see only premium items by filtering prices greater than ₹500 using a custom function passed as a parameter.

- Input: List of prices
- Output: Only prices above ₹500 using custom filter function
- Concept: Higher-order function, function as parameter

### 8. Star Rating System

Manually place 5 clickable stars. On selection, show how many stars the user selected with a message.

- 5 clickable stars (use `onclick`)
- Highlight selected stars and show rating
- Concept: Dynamic DOM, function with parameter

### 9. Typing Animation (Callback)

Simulate typing effect by showing one character at a time. After the whole message is shown, trigger a final message.

- Show message letter by letter
- Call final callback after message completes
- Concept: Callback function, `setTimeout`

### 10. Quiz Score Tracker (Mix of Functions)

Display a simple 3-question multiple choice quiz. On submit, calculate the total score and display feedback.

- Display 3 MCQs with radio options
- On submit, calculate score and show result
- Concept: Function declarations, return, arrow functions, callback for score calculation

Happy coding! Make sure you use comments in your code to explain your logic. Reach out if you get stuck. Good luck!

— Raghavendra B